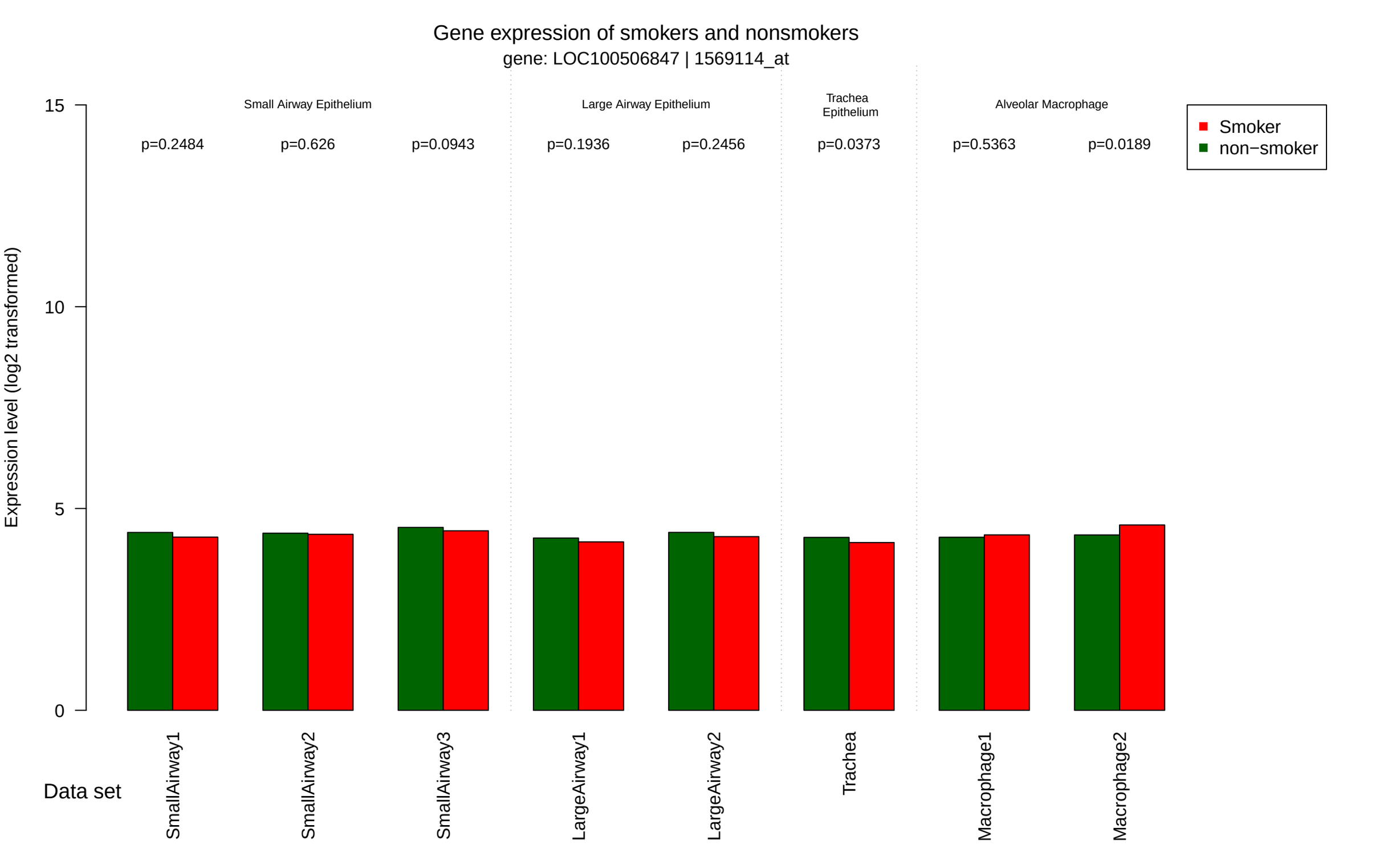
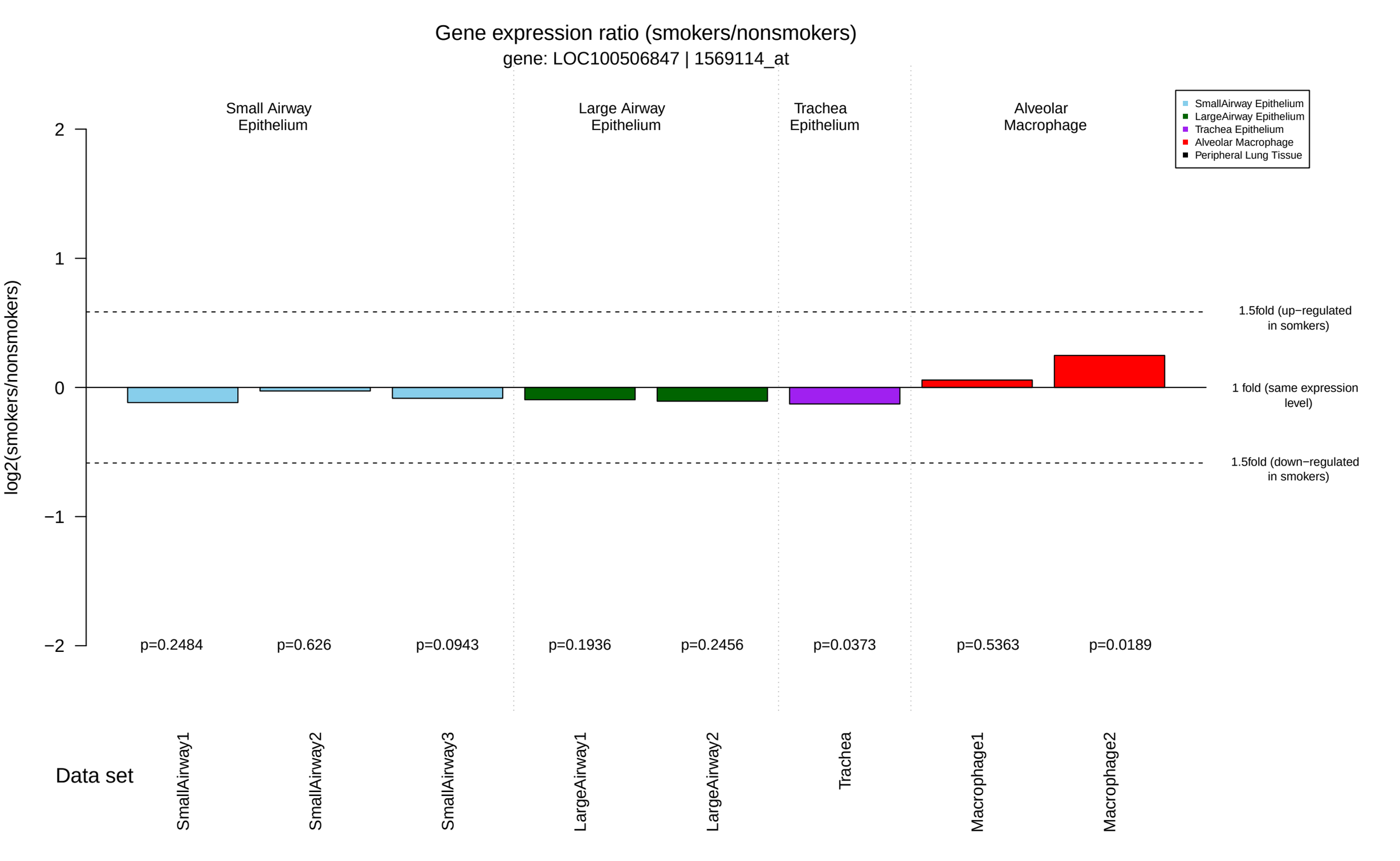
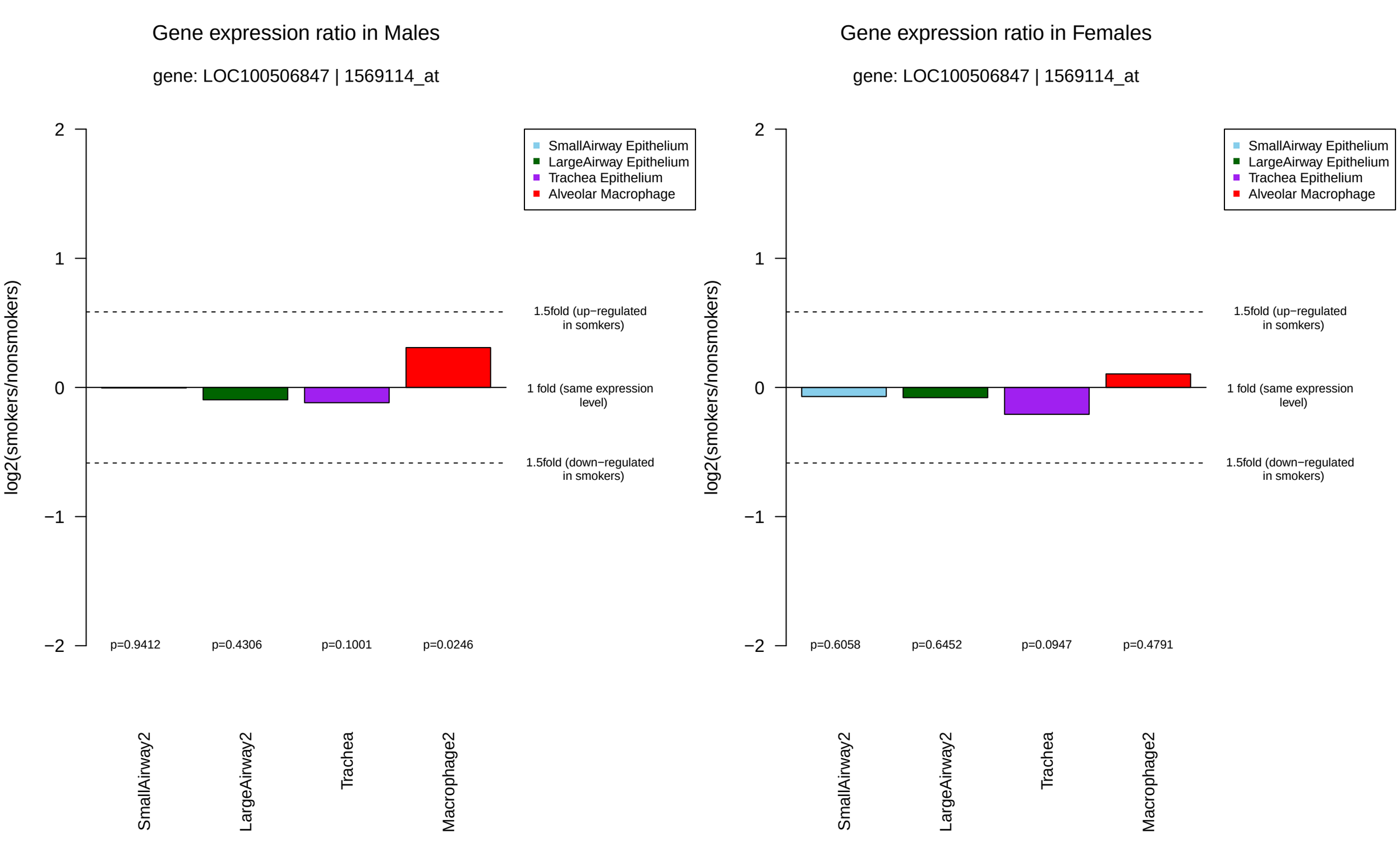
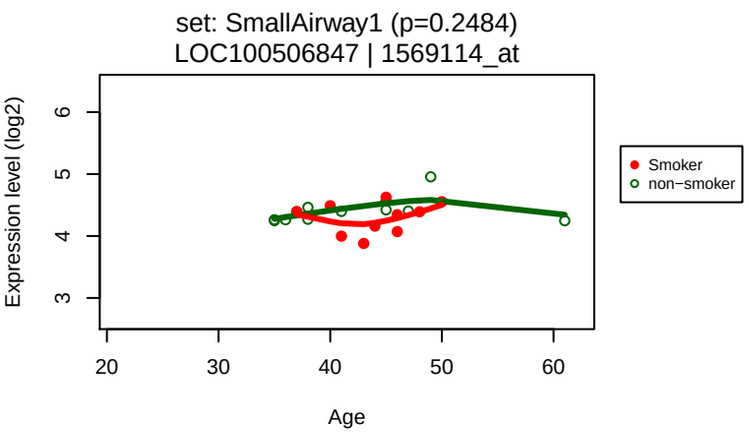
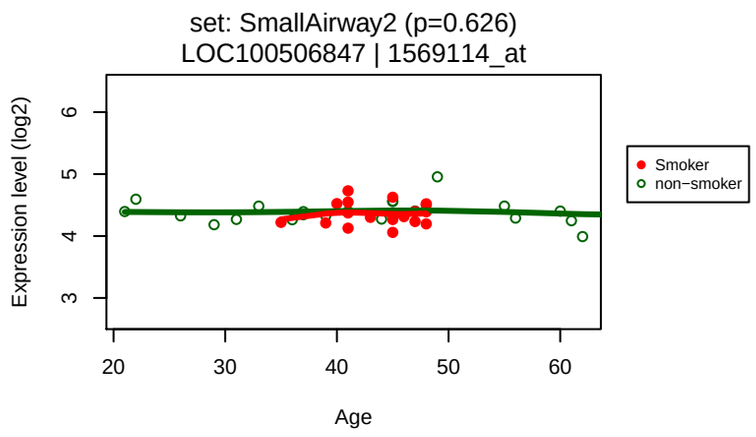
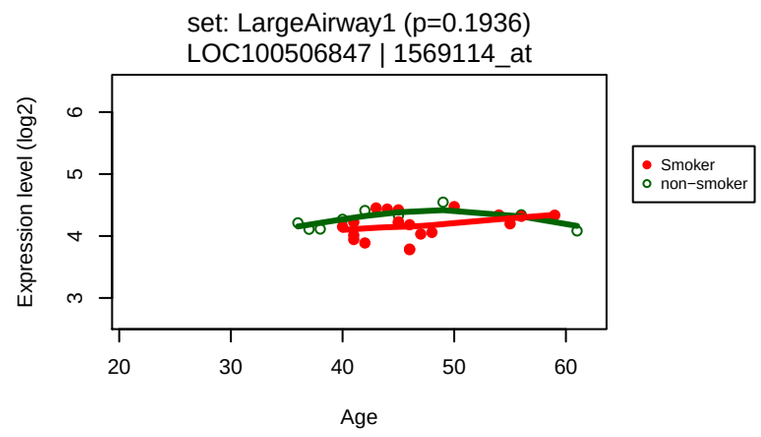

Gene expression of smokers and nonsmokers
gene: LOC100506847 | 1569114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2484
p=0.626
p=0.0943
p=0.1936
p=0.2456
p=0.0373
p=0.5363
p=0.0189
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

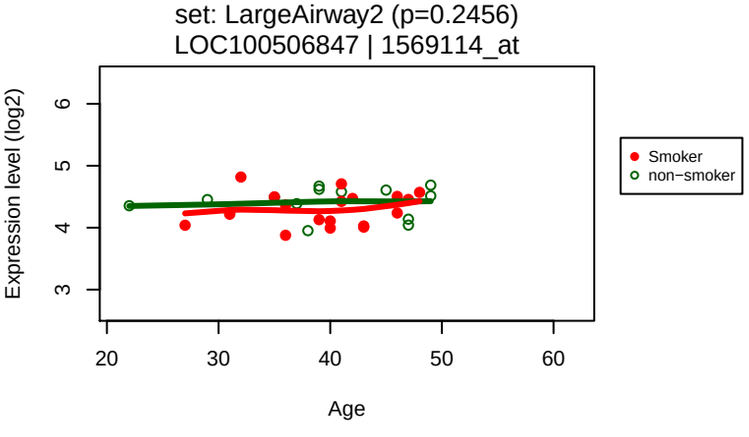

Gene expression ratio (smokers/nonsmokers)
gene: LOC100506847 | 1569114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2484
p=0.626
p=0.0943
p=0.1936
p=0.2456
p=0.0373
p=0.5363
p=0.0189
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

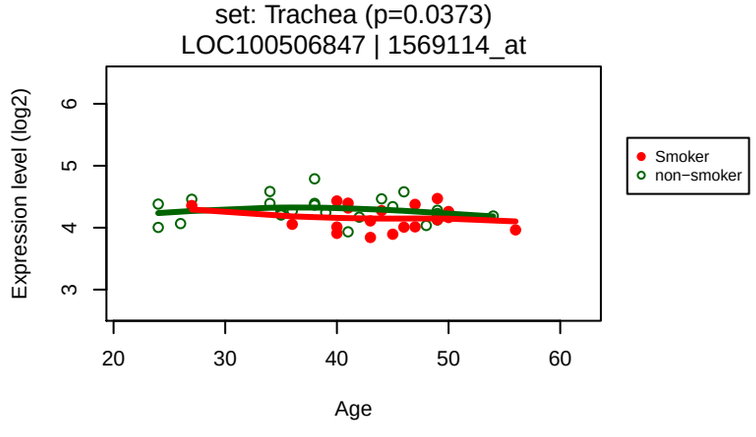

Gene expression ratio in Males
gene: LOC100506847 | 1569114_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9412
p=0.4306
p=0.1001
p=0.0246
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100506847 | 1569114_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6058
p=0.6452
p=0.0947
p=0.4791
SmallAirway2
LargeAirway2
Trachea
Macrophage2

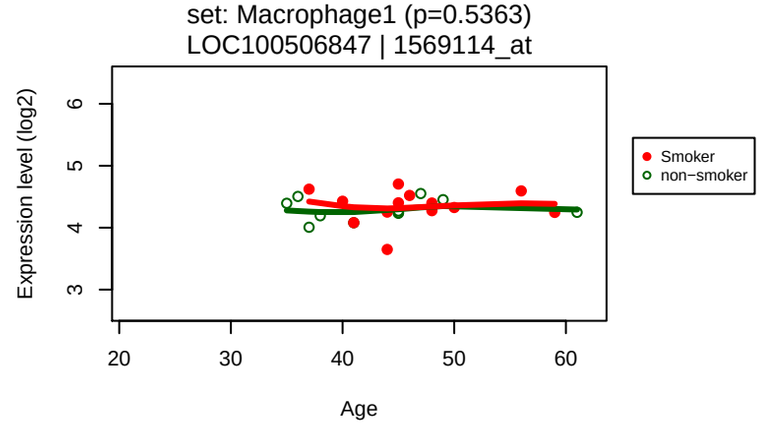

set: SmallAirway1 (p=0.2484)
LOC100506847 | 1569114_at
Expression level (log2)
Age
Smoker
non−smoker

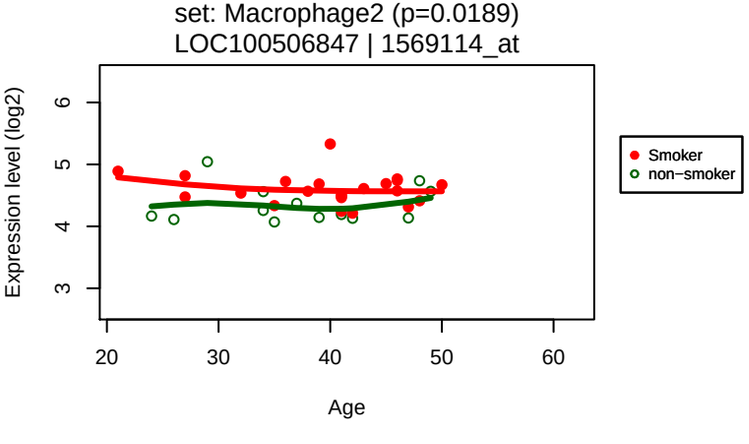

set: SmallAirway2 (p=0.626)
LOC100506847 | 1569114_at
Expression level (log2)
Age
Smoker
non−smoker

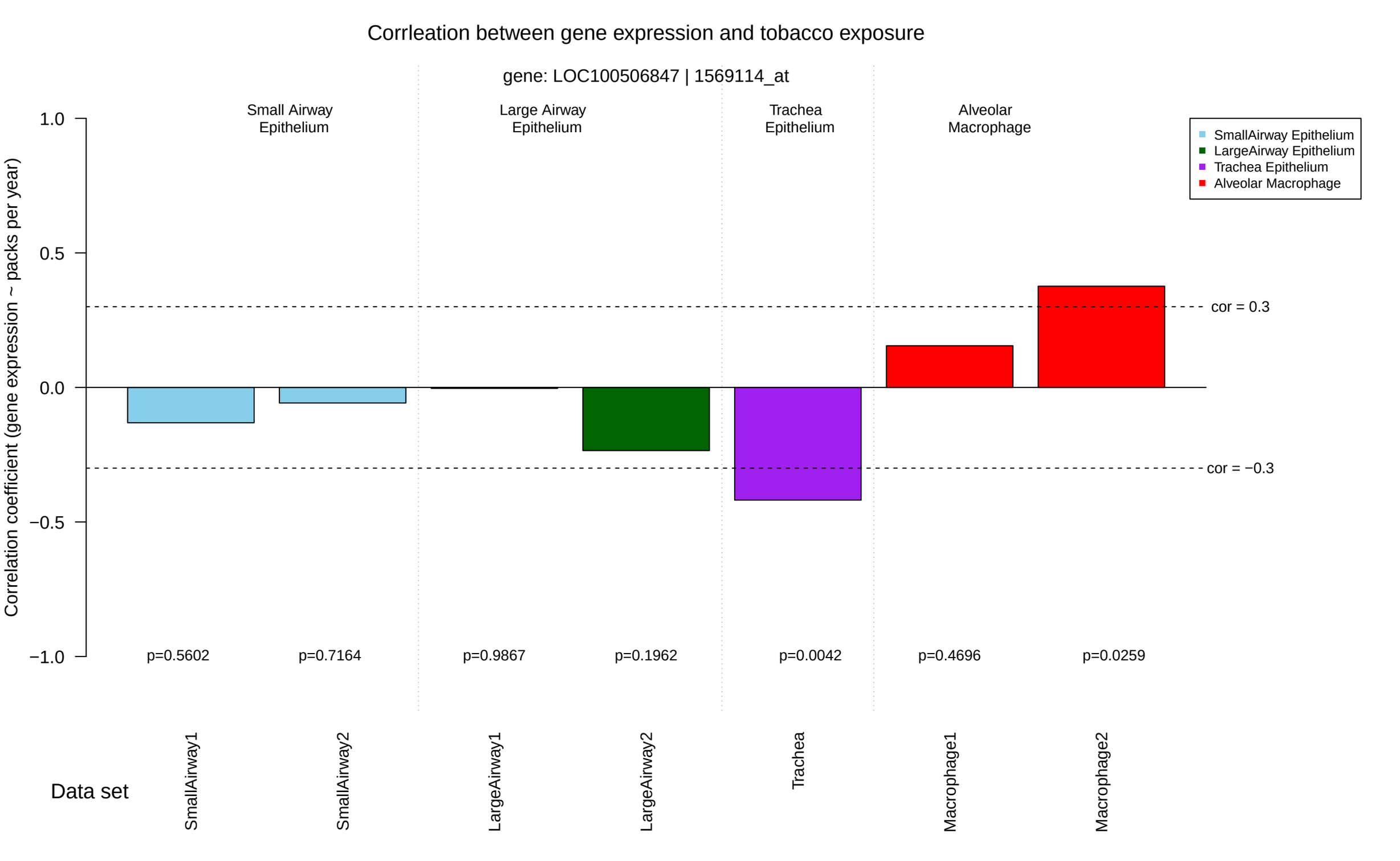

set: LargeAirway1 (p=0.1936)
LOC100506847 | 1569114_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2456)
LOC100506847 | 1569114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0373)
LOC100506847 | 1569114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5363)
LOC100506847 | 1569114_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0189)
LOC100506847 | 1569114_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100506847 | 1569114_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5602
p=0.7164
p=0.9867
p=0.1962
p=0.0042
p=0.4696
p=0.0259
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2